

On the intercluster and inner-cluster distances of a tree

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Abstract

For two vertex clusters of a tree metric, we show that the sum of the average inner-cluster distances is always less than or equal to twice of the average intercluster distance. We show the feature in a more general form of weighted distance. This feature provides a 2-approximation algorithm for the minimum average intercluster distance spanning tree problem, which is a generalization of the minimum routing cost spanning tree or minimum average distance spanning tree problem. Generalization to multiple clusters is also investigated.

1 Introduction

Let $T = (V, E, w)$ be a tree with vertex set V , edge set E , and nonnegative edge length w . For two vertex clusters (subsets) V_1 and V_2 , let $\mu_T(V_1, V_2) = (|V_1||V_2|)^{-1} \sum_{u \in V_1} \sum_{v \in V_2} d_T(u, v)$ denote the average distance between V_1 and V_2 , in which $d_T(u, v)$ denotes the distance between two vertices. For arbitrary V_1 and V_2 , we show that

$$\mu_T(V_1, V_1) + \mu_T(V_2, V_2) \leq 2\mu_T(V_1, V_2),$$

i.e., the sum of the average inner-cluster distances is always less than or equal to twice of the average intercluster distance.

It can be thought of as a feature of tree metrics since the inequality holds for any tree with nonnegative edge length but it is not always true for a graph with cycles. For example, consider a complete bipartite graph G with vertex set $V_1 \cup V_2$ and edges of unit length. Suppose that $|V_1| = |V_2| = n$. It can be easily computed that $\mu_G(V_1, V_1) = \mu_G(V_2, V_2) = 2(n-1)/n$ and $\mu_G(V_1, V_2) = 1$, and we have that $\mu_G(V_1, V_1) + \mu_G(V_2, V_2) > 2\mu_G(V_1, V_2)$ for $n > 2$. In fact, we

show the result in a more general form that there is a vertex weight function and the average distance is the mean of the weighted distance. The case of multiple clusters has also been studied and similar results are derived in this paper.

For a given graph, it is interesting to find a spanning tree with minimum total distance between all pairs of vertices. Such a problem is called the *minimum routing cost spanning tree* (MRCT) problem, or sometimes called the *minimum average distance spanning tree* (MAD) problem. The MRCT problem is NP-hard [7], and several algorithmic results have been reported, including approximation algorithms [8, 12, 15] and exact algorithms [3, 5]. Researchers are also attracted by several variants of the problem, including the weighted total distance [6, 10, 13, 14] and multiple sources [1, 4, 9, 10]. A survey of the MRCT problem and its variants can be found in [11]. Here we consider another variant of the MRCT problem.

PROBLEM: The minimum average inter-cluster distance spanning tree (MAID_k).

INSTANCE: A graph $G = (V, E, w)$ and subsets V_i , $1 \leq i \leq k$, of V .

GOAL: A spanning tree T minimizing $c(T) = \sum_{i < j} \mu_T(V_i, V_j)$.

When $k = 2$ and $V_1 = V_2 = V$, the problem is equivalent to the MRCT problem. Since the MRCT problem is NP-hard, the MAID_k problem is NP-hard even for $k = 2$. Note that the MAID₂ is a generalization of the multiple source MRCT problem, in which a set S of source vertices is given and the objective function is the total distance from all sources to all vertices. The MAID₂ is equivalent to the multiple source MRCT if we let $V_1 = S$ and $V_2 = V$. By the feature of the intercluster distance of a tree metric, we show that there is a shortest-paths tree which approximates the MAID with error ratio 2.

The remaining sections are organized as follows: In Section 2, we show the intercluster distance

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property of two clusters. The 2-approximation of MAID_2 is given in Section 3. Then the results of the multiple clusters and MAID_k for $k > 2$ are shown in Section 4 and 5 respectively. Finally concluding remarks are in Section 6.

2 Distance between two clusters

Let $G = (V, E, w)$ be a graph, in which w is a nonnegative edge length function. For $u, v \in V$, by $d_G(u, v)$, we denote the *distance* between the two vertices on G , which is the total length of the edges in their shortest path. When G is a tree, there is a unique simple path between two vertices, which is also the shortest path. For a graph G , let $V(G)$ denote its vertex set. Let $f : V(G) \rightarrow \mathcal{R}$ be a vertex weight function. For $U \subset V(G)$, we use $f(U)$ to denote the total weight of the vertices in U , i.e., $f(U) = \sum_{v \in U} f(v)$. Also, for any subgraph G' of G , let $f(G')$ denote $f(V(G'))$.

Definition 1: Let G be a graph with vertex set V , and f and h two vertex weight functions. Define $W_G(f, h) = \sum_{u, v \in V} f(u)h(v)d_G(u, v)$.

Theorem 1: Let T be a tree with vertex set V , and f and h two vertex weight functions. Then $H^2W_T(f, f) + F^2W_T(h, h) \leq 2FHW_T(f, h)$, in which $F = f(V)$ and $H = h(V)$.

Define a function

$$\begin{aligned} \Delta(T, f, h) \\ = H^2W_T(f, f) + F^2W_T(h, h) - 2FHW_T(f, h). \end{aligned}$$

In the following, we shall give a proof of Theorem 1 by showing that $\Delta(T, f, h) \leq 0$. First we define the edge contraction of a graph. In general, the contraction of an edge is to remove an edge and combine its two endpoints into a new vertex. Since we consider only trees and the distances on trees, for our convenience, we define the contraction operation as follows:

Definition 2: Let $T = (V, E, w)$ be a tree and $e \in E$. By T/e , we denote a contracted tree which is obtained from T by setting $w(e) = 0$.

Proposition 1: For any $e \in E$, if $T' = T/e$, then $\Delta(T, f, h) \leq \Delta(T', f, h)$.

Proof: Let X and Y be vertex sets of the two subtrees obtained by removing e from T . By definition, for $\{u, v\} \subset X$ or $\{u, v\} \subset Y$, $d_{T'}(u, v) =$

$d_T(u, v)$; and for the case that u and v are in different subsets, we have $d_{T'}(u, v) = d_T(u, v) - w(e)$. Therefore,

$$\begin{aligned} & W_T(f, f) - W_{T'}(f, f) \\ &= \sum_{u \in X} \sum_{v \in Y} f(u)f(v)w(e) \\ & \quad + \sum_{u \in Y} \sum_{v \in X} f(u)f(v)w(e) \\ &= (f(X)f(Y) + f(Y)f(X))w(e) \\ &= 2f(X)f(Y)w(e) \end{aligned}$$

Similarly $W_T(h, h) - W_{T'}(h, h) = 2h(X)h(Y)w(e)$ and $W_T(f, h) - W_{T'}(f, h) = (f(X)h(Y) + f(Y)h(X))w(e)$.

$$\begin{aligned} & (\Delta(T', f, h) - \Delta(T, f, h))/w(e) \\ &= 2HF(f(X)h(Y) + f(Y)h(X)) \\ & \quad - 2H^2f(X)f(Y) - 2F^2h(X)h(Y) \end{aligned}$$

Since

$$\begin{aligned} & HF(f(X)h(Y) + f(Y)h(X)) \\ &= HF(f(X)(H - h(X)) + (F - f(X))h(X)) \\ &= H^2Ff(X) + HF^2h(X) - 2HFf(X)h(X) \end{aligned}$$

and

$$\begin{aligned} & H^2f(X)f(Y) + F^2h(X)h(Y) \\ &= H^2f(X)(F - f(X)) + F^2h(X)(H - h(X)) \\ &= H^2Ff(X) + HF^2h(X) \\ & \quad - (Hf(X))^2 - (Fh(X))^2, \end{aligned}$$

we have that

$$\begin{aligned} & \Delta(T', f, h) - \Delta(T, f, h) \\ &= 2((Hf(X))^2 + (Fh(X))^2 \\ & \quad - 2HFf(X)h(X))w(e) \\ &= 2(Hf(X) - Fh(X))^2w(e) \\ &\geq 0 \end{aligned}$$

□

Proposition 2: $\Delta(T, f, h) \leq 0$.

Proof: In arbitrary order, we contract the edges of T one by one. Let T_i , $1 \leq i \leq |E|$, be the resulted tree. By Proposition 1, we have that $\Delta(T, f, h) \leq \Delta(T_1, f, h)$ and $\Delta(T_i, f, h) \leq \Delta(T_{i+1}, f, h)$ for all i . By definition, the length of each edge of $T_{|E|}$ is zero. Clearly $\Delta(T_{|E|}, f, h) = 0$, and consequently

$$\Delta(T, f, h) \leq 0. \quad \square$$

Theorem 1 directly follows the definition of $\Delta(T, f, h)$ and Proposition 2.

Definition 3: Let G be a graph, V_1 and V_2 subsets of $V(G)$, and f a vertex weight function. The total weighted intercluster distance is defined by $d_G(V_1, V_2; f) = \sum_{u \in V_1} \sum_{v \in V_2} f(u)f(v)d_G(u, v)$, and the average intercluster distance is defined by $\mu_G(V_1, V_2; f) = (f(V_1)f(V_2))^{-1}d_G(V_1, V_2; f)$.

When f is uniform, i.e., $f(v) = 1$ for each vertex v , we shall use $d_G(V_1, V_2)$ for $d_G(V_1, V_2; f)$ and $\mu_G(V_1, V_2)$ for $\mu_G(V_1, V_2; f)$.

Theorem 1 is helpful to show the following properties of the intercluster distances of a tree metric.

Corollary 2: Let T be a tree, V_1 and V_2 subsets of $V(T)$, and δ a vertex weight function. Then $\delta(V_2)^2 d_T(V_1, V_1; \delta) + \delta(V_1)^2 d_T(V_2, V_2; \delta) \leq 2\delta(V_1)\delta(V_2)d_T(V_1, V_2; \delta)$.

Proof: Let $f(v) = \delta(v)$ if $v \in V_1$ and $f(v) = 0$ otherwise, and $h(v) = \delta(v)$ if $v \in V_2$ and $h(v) = 0$ otherwise. The result directly follows Theorem 1. $\square$

By the definition of the average intercluster distance, the above property can be rewritten as follows.

Corollary 3: Let T be a tree, V_1 and V_2 subsets of $V(T)$, and f a vertex weight function. Then $\mu_T(V_1, V_1; f) + \mu_T(V_2, V_2; f) \leq 2\mu_T(V_1, V_2; f)$.

For the unweighted case, we have simpler forms. That is,

$$\begin{aligned} & |V_2|^2 d_T(V_1, V_1) + |V_1|^2 d_T(V_2, V_2) \\ & \leq 2|V_1||V_2|d_T(V_1, V_2) \end{aligned} \quad (1)$$

and

$$\mu_T(V_1, V_1) + \mu_T(V_2, V_2) \leq 2\mu_T(V_1, V_2) \quad (2)$$

The next result shows that the minimum of the inner-cluster distances is always less than or equal to the intercluster distance. But here it is required that the vertex weight function is nonnegative.

Corollary 4: Let T be a tree, V_1 and V_2 subsets of $V(T)$, and f a nonnegative vertex weight function. Then $\min\{d_T(V_1, V_1; f), d_T(V_2, V_2; f)\} \leq d_T(V_1, V_2; f)$.

Proof:

$$\begin{aligned} & \min\{d_T(V_1, V_1; f), d_T(V_2, V_2; f)\} \\ & \leq \frac{1}{f(V_2)^2 + f(V_1)^2} \\ & \quad (f(V_2)^2 d_T(V_1, V_1; f) + f(V_1)^2 d_T(V_2, V_2; f)) \\ & \leq \frac{2f(V_1)f(V_2)}{f(V_2)^2 + f(V_1)^2} d_T(V_1, V_2; f) \\ & \leq d_T(V_1, V_2; f) \end{aligned}$$

The last step is obtained from that $f(V_2)^2 + f(V_1)^2 \geq 2f(V_1)f(V_2)$ since

$$\begin{aligned} & f(V_2)^2 + f(V_1)^2 - 2f(V_1)f(V_2) \\ & = (f(V_1) - f(V_2))^2 \geq 0 \end{aligned}$$

and that f is nonnegative. $\square$

Similarly, for the unweighted case, we have that

$$\min\{d(V_1, V_1), d(V_2, V_2)\} \leq d(V_1, V_2). \quad (3)$$

3 The minimum average intercluster distance spanning tree

In this section, we consider the MAID₂ problem. Given a graph $G = (V, E, w)$ and subsets V_1 and V_2 of V , the goal is to find a spanning tree T minimizing $c(T) = \mu_T(V_1, V_2)$. we shall show the following result.

Theorem 5: For a graph $G = (V, E, w)$ and two vertex subsets V_1 and V_2 , there is a vertex $r \in V_1 \cup V_2$ such that any shortest-paths tree rooted at r is a 2-approximation of the MAID₂. Such a 2-approximation can be computed in $O(pn \log n + pm)$ time, in which $n = |V|$, $m = |E|$, and $p = |V_1 \cup V_2|$.

Proof: Let $T(r)$ be a shortest-paths tree rooted at r . It is easy to see that

$$\begin{aligned} & d_{T(r)}(V_1, V_2) \quad (4) \\ & \leq \sum_{u \in V_1} \sum_{v \in V_2} (d_{T(r)}(r, u) + d_{T(r)}(r, v)) \\ & = \sum_{u \in V_1} \sum_{v \in V_2} (d_G(r, u) + d_G(r, v)) \\ & = \sum_{v \in V_2} d_G(r, V_1) + \sum_{u \in V_1} d_G(r, V_2) \\ & = n_2 d_G(r, V_1) + n_1 d_G(r, V_2), \end{aligned} \quad (5)$$

in which $n_i = |V_i|$ for each i .

Summing up the inequality (5) over all $r \in V_1$, we have that

$$\sum_{r \in V_1} d_{T(r)}(V_1, V_2) \leq n_2 d_G(V_1, V_1) + n_1 d_G(V_1, V_2).$$

Therefore,

$$\begin{aligned} & \min_{r \in V_1} \{d_{T(r)}(V_1, V_2)\} \\ & \leq \frac{1}{n_1} \sum_{r \in V_1} d_{T(r)}(V_1, V_2) \\ & \leq \frac{n_2}{n_1} d_G(V_1, V_1) + d_G(V_1, V_2). \end{aligned}$$

Since $c(T) = (n_1 n_2)^{-1} d_T(V_1, V_2)$ for any spanning tree T ,

$$\begin{aligned} & \min_{r \in V_1} \{c(T(r))\} \\ & \leq n_1^{-2} d_G(V_1, V_1) + (n_1 n_2)^{-1} d_G(V_1, V_2) \\ & = \mu_G(V_1, V_1) + \mu_G(V_1, V_2). \end{aligned} \quad (6)$$

Similarly, by summing up (5) over all $r \in V_2$, we have

$$\sum_{r \in V_2} d_{T(r)}(V_1, V_2) \leq n_2 d_G(V_1, V_2) + n_1 d_G(V_2, V_2),$$

and therefore

$$\min_{r \in V_2} \{c(T(r))\} \leq \mu_G(V_1, V_2) + \mu_G(V_2, V_2). \quad (7)$$

By (6) and (7), we obtain that

$$\begin{aligned} & \min_{r \in V_1 \cup V_2} \{c(T(r))\} \\ & \leq \frac{1}{2} \left(\min_{r \in V_1} \{c(T(r))\} + \min_{r \in V_2} \{c(T(r))\} \right) \\ & \leq \mu_G(V_1, V_2) + \frac{1}{2} (\mu_G(V_1, V_1) + \mu_G(V_2, V_2)) \\ & \leq \mu_T(V_1, V_2) + \frac{1}{2} (\mu_T(V_1, V_1) + \mu_T(V_2, V_2)), \end{aligned}$$

where T is any spanning tree of G . By the inter-cluster distance property (2),

$$\mu_T(V_1, V_1) + \mu_T(V_2, V_2) \leq 2\mu_T(V_1, V_2).$$

Consequently $\min_{r \in V_1 \cup V_2} \{c(T(r))\}$ is at most twice of the optimal. To find a 2-approximation of the MAID₂, we compute the cost for the shortest-paths tree rooted at every vertex in $V_1 \cup V_2$ and choose the best one. Since a shortest-path tree can be constructed in $O(n \log n + m)$ time [2], the time complexity of such an algorithm is $O(pn \log n + pm)$. $\square$

4 Multi-clusters

For a graph with vertex set V and two vertex weight functions f and h , let

$$\alpha_G(f, h) = (f(V)h(V))^{-1} W_G(f, h),$$

in which $W_G(f, h) = \sum_{u, v \in V} f(u)h(v)d(u, v)$ is defined previously.

Theorem 6: Let T be a tree and f_i , $1 \leq i \leq k$, be k vertex weight functions defined on $V(T)$. Then $(k-1) \sum_{i=1}^k \alpha_T(f_i, f_i) \leq 2 \sum_{i < j} \alpha_T(f_i, f_j)$.

Proof: Let $F_i = f(V_i)$ for each i . By Theorem 1, we have

$$F_j^2 W_T(f_i, f_i) + F_i^2 W_T(f_j, f_j) \leq 2F_i F_j W_T(f_i, f_j),$$

and it is implied that

$$\alpha_T(f_i, f_i) + \alpha_T(f_j, f_j) \leq 2\alpha_T(f_i, f_j). \quad (8)$$

Summing up (8) for $1 \leq i < j \leq k$, we obtain the result. $\square$

In Theorem 6, for each i , let $f_i(v) = f(v)$ for $v \in V_i$ and 0 for otherwise. We obtain the following result. Recall that $\mu_T(V_i, V_j; f) = d_T(V_i, V_j; f) / (f(V_i)f(V_j))$ is defined in Definition 3.

Corollary 7: Let T be a tree and f be a vertex weight function. If V_i , $1 \leq i \leq k$, are subsets of $V(T)$, then $(k-1) \sum_{i=1}^k \mu_T(V_i, V_i; f) \leq 2 \sum_{i < j} \mu_T(V_i, V_j; f)$.

In particular, if $f(v) = 1$ for $v \in V$, we obtain the next corollary.

Corollary 8: Let T be a tree. If V_i , $1 \leq i \leq k$, are subsets of $V(T)$, then $(k-1) \sum_{i=1}^k \mu_T(V_i, V_i) \leq 2 \sum_{i < j} \mu_T(V_i, V_j)$.

Furthermore, by Corollary 7, since

$$\min_i \{\mu_T(V_i, V_i; f)\} \leq \frac{1}{k} \sum_{i=1}^k \mu_T(V_i, V_i; f),$$

we have the next result.

$$\min_i \{\mu_T(V_i, V_i; f)\} \leq \left(\frac{k}{2}\right)^{-1} \sum_{i < j} \mu_T(V_i, V_j; f) \quad (9)$$

Note that the right-hand side is the average of $\mu_T(V_i, V_j; f)$.

5 MAID_k for $k > 2$

We turn to the MAID_k problem for $k > 2$. We shall also derive a polynomial time 2-approximation algorithm.

Theorem 9: For a graph $G = (V, E, w)$ and k subsets V_i , $1 \leq i \leq k$ of V , there is a vertex $r \in \bigcup_i V_i$ such that any shortest-paths tree rooted at r is a 2-approximation of the MAID_k. Such a 2-approximation can be computed in $O(pn \log n + pm)$ time, in which $n = |V|$, $m = |E|$, and $p = |\bigcup_i V_i|$.

Proof:

Let $T(r)$ be a shortest-paths tree rooted at r . Let $n_i = |V_i|$. Similar to the case of $k = 2$, it is easy to see that, for $1 \leq i, j \leq k$,

$$d_{T(r)}(V_i, V_j) \leq n_j \sum_{v \in V_i} d_G(r, v) + n_i \sum_{v \in V_j} d_G(r, v).$$

That is,

$$\mu_{T(r)}(V_i, V_j) \leq \mu_G(r, V_i) + \mu_G(r, V_j).$$

Summing up the inequality for all $1 \leq i < j \leq k$, we have that

$$c(T(r)) \leq (k-1) \sum_i \mu_G(r, V_i). \quad (10)$$

For any j , summing (10) for all $r \in V_j$ and taking the mean, we obtain

$$\min_{r \in V_j} \{c(T(r))\} \leq (k-1) \sum_i \mu_G(V_i, V_j). \quad (11)$$

By (11),

$$\begin{aligned} & \min_{r \in V_j, \forall j} \{c(T(r))\} \\ & \leq \frac{1}{k} \sum_j \left((k-1) \sum_i \mu_G(V_i, V_j) \right) \\ & = \frac{(k-1)}{k} \sum_{i,j} \mu_G(V_i, V_j) \\ & \leq \frac{(k-1)}{k} \left(\sum_i \mu_T(V_i, V_i) + \sum_{i \neq j} \mu_T(V_i, V_j) \right), \end{aligned}$$

where T is any spanning tree of G . By Corollary 8, $(k-1) \sum_i \mu_T(V_i, V_i) \leq 2 \sum_{i < j} \mu_T(V_i, V_j)$. Then

$$\begin{aligned} & \min_{r \in V_j, \forall j} \{c(T(r))\} \\ & \leq \frac{(k-1)}{k} \left(\frac{2}{k-1} \sum_{i < j} \mu_T(V_i, V_j) \right) \end{aligned}$$

$$\begin{aligned} & + 2 \sum_{i < j} \mu_T(V_i, V_j) \Big) \\ & = 2 \sum_{i < j} \mu_T(V_i, V_j) \end{aligned}$$

We have show that there is a vertex $r \in \bigcup_i V_i$ such that any shortest-paths tree rooted at r is a 2-approximation of the MAID_k. To find a 2-approximation, we compute the cost for the shortest-paths tree rooted at every vertex in $\bigcup_i V_i$ and choose the best one. The time complexity of such an algorithm is $O(pn \log n + pm)$. $\square$

6 Concluding remarks

In this paper, we show a property of the average intercluster distance of a tree metric, and it can be applied to find a 2-approximation of the minimum average intercluster distance spanning tree. Applications to other problems are expected. The MAID problem generalizes the multiple source MRCT problem and is also interesting. Developing better approximation algorithms will be an interesting future work.

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